A. Informasi & Ringkasan

1. Identitas

NIM	11S25038
Nama	EVAN BENTA PURBA
Kelas	11IF-2
Judul Praktikum	Tipe Bentukan : Date

2. Capaian & Ringkasan Pemahaman

Hasil Capaian setelah mengikuti praktikum:

- A (Penyelesaian): Selesai (1), Tidak Selesai (0).
- B (Pemahaman): Tidak Paham (1), Kurang Paham (2), Cukup Paham (3), Paham (4), Sangat Paham (5).

no	Kriteria	A	B
1	Mengimplementasikan Tipe Bentukan Date	1	3

3.Foto Praktikan



B. Penilaian Observasi

Setiap Mahasiswa yang mengikuti kegiatan praktikum perlu melakukan penilaian terhadap diri sendiri dan rekan kerja berdasarkan [Indikator Penilaian Observasi Praktikum].

1. Penilaian Terhadap Diri Sendiri

No	Indikator	Skala (1 - 5)
1	Kepatuhan Terhadap Prosedur	3

2	Kemampuan Penyelesaian Tugas	4
3	Kerja Sama Tim	4
4	Inisiatif dan Problem Solving	5
5	Kedisiplinan dan Etika Kerja	5

2. Penilaian Terhadap Rekan Kerja

Harus melakukan penilaian terhadap rekan kerja selama aktivitas praktikum berlangsung.

a) Judika Ompusunggu (11S25044)

No	Indikator	Skala (1 - 5)
1	Kepatuhan Terhadap Prosedur	5
2	Kemampuan Penyelesaian Tugas	3
3	Kerja Sama Tim	5
4	Inisiatif dan Problem Solving	4
5	Kedisiplinan dan Etika Kerja	5

b) Alfredo sianturi (11S25023)

No	Indikator	Skala (1 - 5)
1	Kepatuhan Terhadap Prosedur	5
2	Kemampuan Penyelesaian Tugas	3
3	Kerja Sama Tim	5
4	Inisiatif dan Problem Solving	4
5	Kedisiplinan dan Etika Kerja	5

c) Deon Hutagalung (11S25039)

No	Indikator	Skala (1 - 5)
1	Kepatuhan Terhadap Prosedur	3
2	Kemampuan Penyelesaian Tugas	4
3	Kerja Sama Tim	2
4	Inisiatif dan Problem Solving	3
5	Kedisiplinan dan Etika Kerja	4

C. Laporan Aktivitas Praktikum

1. Mengimplementasikan Tipe Bentukan date

Buatlah tipe bentukan Date, tipe ini terdiri dari hari (dd), bulan (mm), dan tahun (yy).

1. Konstruktor

a) makeDate(dd, mm, yy), untuk membentuk Date, dari hari dd,

bulan mm, dan tahun yy.

2. Selektor

a) day(d), untuk mengembalikan nilai hari (dd) dari Date d.

b) month(d), untuk mengembalikan nilai bulan (mm) dari Date d.

c) year(d), untuk mengembalikan nilai tahun (yy) dari Date d.

3. Operator

a) nextDay(d), untuk menentukan tanggal berikutnya dari Date d. Harus diperhatikan jumlah hari pada setiap bulan dan juga memperhatikan tahun kabisat. contoh test case yang harus diuji adalah:

Test case	Input (dd mm yy)	Output (dd mm yy)	Keterangan
1	21 4 2006	22 4 2006	penambahan biasa
2	15 2 2016	16 2 2016	penambahan biasa
3	28 2 2016	29 2 2016	tahun kabisat
4	28 2 2015	1 3 2015	bukan tahun kabisat
5	31 12 2015	1 1 2016	akhir tahun
6	31 1 2016	1 2 2016	bulan dengan 31 hari
7	30 11 2016	1 12 2016	bulan dengan 30 hari

b) yesterday(d), untuk menentukan tanggal sebelumnya dari d.

c) addDay(d, n), untuk menambahkan hari sebanyak n pada tanggal d.

d) subDay(d, n), untuk mengurangi hari sebanyak n pada tanggal d.

e) nextMonth(d), untuk menentukan bulan berikutnya dari tanggal d.

f) prevMonth(d), untuk menentukan bulan sebelumnya dari tanggal d.

g) addMonth(d, n), untuk menambahkan bulan sebanyak n pada tanggal d.

h) subMonth(d, n), untuk mengurangi bulan sebanyak n pada tanggal d.

i) addYear(d, n), untuk menambahkan tahun sebanyak n pada tanggal d.

j) subYear(d, n), untuk mengurangi tahun sebanyak n pada tanggal d.

Kriteria tahun kabisat:

- Jika angka tahun itu habis dibagi 400, maka tahun itu sudah pasti tahun kabisat.
- Jika angka tahun itu tidak habis dibagi 400 tetapi habis dibagi 100, maka tahun itu sudah pasti bukan merupakan tahun kabisat.
- Jika angka tahun itu tidak habis dibagi 400, tidak habis dibagi 100 akan tetapi habis dibagi 4, maka tahun itu merupakan tahun kabisat.
- Jika angka tahun tidak habis dibagi 400, tidak habis dibagi 100, dan tidak habis dibagi 4, maka tahun tersebut bukan merupakan tahun kabisat.

4. Predikat (operator relasional Date)

a) isEqD(d1, d2), untuk membandingkan, apakah kedua Date sama.

b) isNotEqD(d1, d2), untuk membandingkan, apakah kedua Date tidak sama.

c) isBefore(d1, d2), mengembalikan nilai benar jika tanggal d1 sebelum d2.

d) isNext(d1, d2), mengembalikan nilai benar jika tanggal d1 sesudah d2.

e) isBeforeN(d1, d2, n), mengembalikan nilai benar jika tanggal d1 sebelum d2 sebanyak n.

f) isNextN(d1, d2, n), mengembalikan nilai benar jika tanggal d1 sesudah d2 sebanyak n.

Hasil yang diharapkan sebagai berikut:

Input	Expect Output
30 4 2014 23 12 2003 2	d1 = Date {dd = 30, mm = 4, yy = 2014} d2 = Date {dd = 23, mm = 12, yy = 2003} n = 2 nextDay(d1) = Date {dd = 1, mm = 5, yy = 2014} nextDay(d2) = Date {dd = 24, mm = 12, yy = 2003} yesterday(d1) = Date {dd = 29, mm = 4, yy = 2014} yesterday(d2) = Date {dd = 22, mm = 12, yy = 2003} addDay(d1, n) = Date {dd = 2, mm = 5, yy = 2014} addDay(d2, n) = Date {dd = 25, mm = 12, yy = 2003} subDay(d1, n) = Date {dd = 28, mm = 4, yy = 2014} subDay(d2, n) = Date {dd = 21, mm = 12, yy = 2003} nextMonth(d1) = Date {dd = 30, mm = 5, yy = 2014} nextMonth(d2) = Date {dd = 23, mm = 1, yy = 2004} prevMonth(d1) = Date {dd = 30, mm = 3, yy = 2014} prevMonth(d2) = Date {dd = 23, mm = 11, yy = 2003} addMonth(d1, n) = Date {dd = 30, mm = 6, yy = 2014} addMonth(d2, n) = Date {dd = 23, mm = 2, yy = 2004} subMonth(d1, n) = Date {dd = 28, mm = 2, yy = 2014} subMonth(d2, n) = Date {dd = 23, mm = 10, yy = 2003} addYear(d1, n) = Date {dd = 30, mm = 4, yy = 2016} addYear(d2, n) = Date {dd = 23, mm = 12, yy = 2005} subYear(d1, n) = Date {dd = 30, mm = 4, yy = 2012} subYear(d2, n) = Date {dd = 23, mm = 12, yy = 2001} isEqD(d1, d2) = False isNotEqD(d1, d2) = True isBefore(d1, d2) = False isNext(d1, d2) = True isBeforeN(d1, d2, n) = False isNextN(d1, d2, n) = False kabisat(yy1) = False kabisat(yy2) = False

```
1       d1 = Date {dd = 1, mm = 1, yy = 2000}
1       d2 = Date {dd = 31, mm = 8, yy = 2002}
2000    n = 3
31      nextDay(d1) = Date {dd = 2, mm = 1, yy = 2000}
8       nextDay(d2) = Date {dd = 1, mm = 9, yy = 2002}
2002    yesterday(d1) = Date {dd = 31, mm = 12, yy = 1999}
3       yesterday(d2) = Date {dd = 30, mm = 8, yy = 2002}
        addDay(d1, n) = Date {dd = 4, mm = 1, yy = 2000}
        addDay(d2, n) = Date {dd = 3, mm = 9, yy = 2002}
        subDay(d1, n) = Date {dd = 29, mm = 12, yy = 1999}
        subDay(d2, n) = Date {dd = 28, mm = 8, yy = 2002}
        nextMonth(d1) = Date {dd = 1, mm = 2, yy = 2000}
        nextMonth(d2) = Date {dd = 30, mm = 9, yy = 2002}
        prevMonth(d1) = Date {dd = 1, mm = 12, yy = 1999}
        prevMonth(d2) = Date {dd = 31, mm = 7, yy = 2002}
        addMonth(d1, n) = Date {dd = 1, mm = 4, yy = 2000}
        addMonth(d2, n) = Date {dd = 30, mm = 11, yy = 2002}
        subMonth(d1, n) = Date {dd = 1, mm = 10, yy = 1999}
        subMonth(d2, n) = Date {dd = 30, mm = 5, yy = 2002}
        addYear(d1, n) = Date {dd = 1, mm = 1, yy = 2003}
        addYear(d2, n) = Date {dd = 31, mm = 8, yy = 2005}
        subYear(d1, n) = Date {dd = 1, mm = 1, yy = 1997}
        subYear(d2, n) = Date {dd = 31, mm = 8, yy = 1999}
        isEqD(d1, d2) = False
        isNotEqD(d1, d2) = True
        isBefore(d1, d2) = False
        isNext(d1, d2) = False
        isBeforeN(d1, d2, n) = False
        isNextN(d1, d2, n) = False
        kabisat(yy1) = True
        kabisat(yy2) = False
```

Code :

tipeDate.hs

```haskell
-- Tipe bentukan Date
data Date = Date { dd :: Int, mm :: Int, yy :: Int } deriving (Show)

-- 1. KONSTRUKTOR
makeDate :: Int -> Int -> Int -> Date
makeDate dd mm yy = Date { dd = dd, mm = mm, yy = yy }

-- 2. SELEKTOR
day :: Date -> Int
day (Date dd _ _) = dd

month :: Date -> Int
month (Date _ mm _) = mm

year :: Date -> Int
year (Date _ _ yy) = yy

isLeapYear :: Int -> Bool
isLeapYear year
    | year `mod` 400 == 0 = True
    | year `mod` 100 == 0 = False
    | year `mod` 4 == 0   = True
    | otherwise           = False

daysInMonth :: Int -> Int -> Int
daysInMonth month year
    | month == 2 && isLeapYear year = 29
    | month == 2 = 28
    | month `elem` [4,6,9,11] = 30
    | otherwise = 31

-- 3. OPERATOR
nextDay :: Date -> Date
nextDay (Date dd mm yy)
    | dd < daysInMonth mm yy = Date (dd + 1) mm yy
    | mm < 12 = Date 1 (mm + 1) yy
    | otherwise = Date 1 1 (yy + 1)

yesterday :: Date -> Date
yesterday (Date dd mm yy)
    | dd > 1 = Date (dd - 1) mm yy
    | mm > 1 = Date (daysInMonth (mm - 1) yy) (mm - 1) yy
    | otherwise = Date 31 12 (yy - 1)

addDay :: Date -> Int -> Date
addDay date 0 = date
addDay date n
    | n > 0 = addDay (nextDay date) (n - 1)
    | n < 0 = addDay (yesterday date) (n + 1)

```

tipeDate.hs

```haskell
addDay date n


subDay :: Date -> Int -> Date
subDay date n = addDay date (-n)

nextMonth :: Date -> Date
nextMonth (Date dd mm yy)
    | mm < 12 = Date (min dd (daysInMonth (mm + 1) yy)) (mm + 1) yy
    | otherwise = Date (min dd (daysInMonth 1 (yy + 1))) 1 (yy + 1)

prevMonth :: Date -> Date
prevMonth (Date dd mm yy)
    | mm > 1 = Date (min dd (daysInMonth (mm - 1) yy)) (mm - 1) yy
    | otherwise = Date (min dd (daysInMonth 12 (yy - 1))) 12 (yy - 1)

addMonth :: Date -> Int -> Date
addMonth date n
    | n == 0 = date
    | n > 0 = addMonth (nextMonth date) (n - 1)
    | n < 0 = addMonth (prevMonth date) (n + 1)

subMonth :: Date -> Int -> Date
subMonth date n = addMonth date (-n)

addYear :: Date -> Int -> Date
addYear (Date dd mm yy) n
    | mm == 2 && dd == 29 && not (isLeapYear (yy + n)) = Date 28 mm (yy + n)
    | otherwise = Date dd mm (yy + n)

subYear :: Date -> Int -> Date
subYear date n = addYear date (-n)

-- 4. PREDIKAT
isEqD :: Date -> Date -> Bool
isEqD (Date dd1 mm1 yy1) (Date dd2 mm2 yy2) =
    dd1 == dd2 && mm1 == mm2 && yy1 == yy2

isNotEqD :: Date -> Date -> Bool
isNotEqD d1 d2 = not (isEqD d1 d2)

isBefore :: Date -> Date -> Bool
isBefore (Date dd1 mm1 yy1) (Date dd2 mm2 yy2)
    | yy1 < yy2 = True
    | yy1 > yy2 = False
    | mm1 < mm2 = True
    | mm1 > mm2 = False
    | dd1 < dd2 = True
    | otherwise = False

```

```haskell

isNext :: Date -> Date -> Bool
isNext d1 d2 = isBefore d2 d1

isBeforeN :: Date -> Date -> Int -> Bool
isBeforeN d1 d2 n = isEqD d1 (addDay d2 (-n))

isNextN :: Date -> Date -> Int -> Bool
isNextN d1 d2 n = isEqD d1 (addDay d2 n)

-- Main function
main :: IO ()
main = do
    dd1 <- readLn
    mm1 <- readLn
    yy1 <- readLn
    dd2 <- readLn
    mm2 <- readLn
    yy2 <- readLn
    n <- readLn

    let d1 = makeDate dd1 mm1 yy1
    let d2 = makeDate dd2 mm2 yy2

    putStrLn $ "d1 = " ++ show d1
    putStrLn $ "d2 = " ++ show d2
    putStrLn $ "n = " ++ show n
    putStrLn $ "nextDay(d1) = " ++ show (nextDay d1)
    putStrLn $ "nextDay(d2) = " ++ show (nextDay d2)
    putStrLn $ "yesterday(d1) = " ++ show (yesterday d1)
    putStrLn $ "yesterday(d2) = " ++ show (yesterday d2)
    putStrLn $ "addDay(d1, n) = " ++ show (addDay d1 n)
    putStrLn $ "addDay(d2, n) = " ++ show (addDay d2 n)
    putStrLn $ "subDay(d1, n) = " ++ show (subDay d1 n)
    putStrLn $ "subDay(d2, n) = " ++ show (subDay d2 n)
    putStrLn $ "nextMonth(d1) = " ++ show (nextMonth d1)
    putStrLn $ "nextMonth(d2) = " ++ show (nextMonth d2)
    putStrLn $ "prevMonth(d1) = " ++ show (prevMonth d1)
    putStrLn $ "prevMonth(d2) = " ++ show (prevMonth d2)
    putStrLn $ "addMonth(d1, n) = " ++ show (addMonth d1 n)
    putStrLn $ "addMonth(d2, n) = " ++ show (addMonth d2 n)
    putStrLn $ "subMonth(d1, n) = " ++ show (subMonth d1 n)
    putStrLn $ "subMonth(d2, n) = " ++ show (subMonth d2 n)
    putStrLn $ "addYear(d1, n) = " ++ show (addYear d1 n)
    putStrLn $ "addYear(d2, n) = " ++ show (addYear d2 n)
    putStrLn $ "subYear(d1, n) = " ++ show (subYear d1 n)
    putStrLn $ "subYear(d2, n) = " ++ show (subYear d2 n)
    putStrLn $ "isEqD(d1, d2) = " ++ show (isEqD d1 d2)
    putStrLn $ "isNotEqD(d1, d2) = " ++ show (isNotEqD d1 d2)
    putStrLn $ "isBefore(d1, d2) = " ++ show (isBefore d1 d2)
    putStrLn $ "isNext(d1, d2) = " ++ show (isNext d1 d2)
    putStrLn $ "isBeforeN(d1, d2, n) = " ++ show (isBeforeN d1 d2 n)
    putStrLn $ "isNextN(d1, d2, n) = " ++ show (isNextN d1 d2 n)
    putStrLn $ "kabisat(yy1) = " ++ show (isLeapYear (yy d1))
    putStrLn $ "kabisat(yy2) = " ++ show (isLeapYear (yy d2))
```

Terminal:

Command setup:

```
ghc -o tipeDate tipeDate.hs
```

```
C:\file kuliah ;p\pemfung\pemfunglagi\pemfung praktikum\ifsS25038-pemfung-praktikum-7>ghc -o tipeDate tipeDate.hs
[1 of 2] Compiling Main             ( tipeDate.hs, tipeDate.o )
[2 of 2] Linking tipeDate.exe
```

command run:

```
tipeDate.exe
```

```
C:\file kuliah ;p\pemfung\pemfunglagi\pemfung praktikum\ifsS25038-pemfung-praktikum-7>
30
4
2014
23
12
2003
2
d1 = Date {dd = 30, mm = 4, yy = 2014}
d2 = Date {dd = 23, mm = 12, yy = 2003}
n = 2
nextDay(d1) = Date {dd = 1, mm = 5, yy = 2014}
nextDay(d2) = Date {dd = 24, mm = 12, yy = 2003}
yesterday(d1) = Date {dd = 29, mm = 4, yy = 2014}
yesterday(d2) = Date {dd = 22, mm = 12, yy = 2003}
addDay(d1, n) = Date {dd = 2, mm = 5, yy = 2014}
addDay(d2, n) = Date {dd = 25, mm = 12, yy = 2003}
subDay(d1, n) = Date {dd = 28, mm = 4, yy = 2014}
subDay(d2, n) = Date {dd = 21, mm = 12, yy = 2003}
nextMonth(d1) = Date {dd = 30, mm = 5, yy = 2014}
nextMonth(d2) = Date {dd = 23, mm = 1, yy = 2004}
prevMonth(d1) = Date {dd = 30, mm = 3, yy = 2014}
prevMonth(d2) = Date {dd = 23, mm = 11, yy = 2003}
addMonth(d1, n) = Date {dd = 30, mm = 6, yy = 2014}
addMonth(d2, n) = Date {dd = 23, mm = 2, yy = 2004}
subMonth(d1, n) = Date {dd = 28, mm = 2, yy = 2014}
subMonth(d2, n) = Date {dd = 23, mm = 10, yy = 2003}
addYear(d1, n) = Date {dd = 30, mm = 4, yy = 2016}
addYear(d2, n) = Date {dd = 23, mm = 12, yy = 2005}
subYear(d1, n) = Date {dd = 30, mm = 4, yy = 2012}
subYear(d2, n) = Date {dd = 23, mm = 12, yy = 2001}
isEqD(d1, d2) = False
isNotEqD(d1, d2) = True
isBefore(d1, d2) = False
isNext(d1, d2) = True
isBeforeN(d1, d2, n) = False
isNextN(d1, d2, n) = False
kabisat(yy1) = False
kabisat(yy2) = False
```

```
C:\file kuliah ;p\pemfung\pemfunglagi\pemfung praktikum\ifsS25038-pemfung-praktikum-7>tipeDate.exe
1
1
2000
31
8
2002
3
d1 = Date {dd = 1, mm = 1, yy = 2000}
d2 = Date {dd = 31, mm = 8, yy = 2002}
n = 3
nextDay(d1) = Date {dd = 2, mm = 1, yy = 2000}
nextDay(d2) = Date {dd = 1, mm = 9, yy = 2002}
yesterday(d1) = Date {dd = 31, mm = 12, yy = 1999}
yesterday(d2) = Date {dd = 30, mm = 8, yy = 2002}
addDay(d1, n) = Date {dd = 4, mm = 1, yy = 2000}
addDay(d2, n) = Date {dd = 3, mm = 9, yy = 2002}
subDay(d1, n) = Date {dd = 29, mm = 12, yy = 1999}
subDay(d2, n) = Date {dd = 28, mm = 8, yy = 2002}
nextMonth(d1) = Date {dd = 1, mm = 2, yy = 2000}
nextMonth(d2) = Date {dd = 30, mm = 9, yy = 2002}
prevMonth(d1) = Date {dd = 1, mm = 12, yy = 1999}
prevMonth(d2) = Date {dd = 31, mm = 7, yy = 2002}
addMonth(d1, n) = Date {dd = 1, mm = 4, yy = 2000}
addMonth(d2, n) = Date {dd = 30, mm = 11, yy = 2002}
subMonth(d1, n) = Date {dd = 1, mm = 10, yy = 1999}
subMonth(d2, n) = Date {dd = 30, mm = 5, yy = 2002}
addYear(d1, n) = Date {dd = 1, mm = 1, yy = 2003}
addYear(d2, n) = Date {dd = 31, mm = 8, yy = 2005}
subYear(d1, n) = Date {dd = 1, mm = 1, yy = 1997}
subYear(d2, n) = Date {dd = 31, mm = 8, yy = 1999}
isEqD(d1, d2) = False
isNotEqD(d1, d2) = True
isBefore(d1, d2) = True
isNext(d1, d2) = False
isBeforeN(d1, d2, n) = False
isNextN(d1, d2, n) = False
kabisat(yy1) = True
kabisat(yy2) = False
```